

Order Delivery & Logistics Performance Analysis

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- **Dataset 1:** 10,000 rows × 21 columns, no missing values, no duplicate rows.
- **Dataset 2:** 10,000 rows × 21 columns, no duplicate rows, but 11,624 missing cells.
- The main business KPIs are **identical in both datasets**, so the missing values in Dataset 2 do not affect the core revenue/order KPIs.

1. Final Business Insights

Executive Business Insights

1. Strong order volume

The analysis covers **10,000 orders**, providing a substantial base for evaluating order fulfillment and delivery performance.

2. Revenue performance

The business generated **₹251,412,079 in total order value** with an **average order value of ₹25,141.21**. This provides a strong baseline for monitoring sales performance.

3. Delivery performance

Out of 10,000 orders:

- **7,472 orders (74.72%)** were On Time.
- **1,716 orders (17.16%)** were delayed.
- **812 orders (8.12%)** were cancelled.

This means the biggest operational opportunity is reducing the delayed and cancelled orders.

4. Logistics cost

Total shipping cost is **₹2,758,648**, with an average shipping cost of **₹275.86 per order**. Monitoring this KPI alongside revenue can help control fulfillment expenses.

5. Regional performance

The **East region** generated the highest revenue at **₹51,443,037**, representing approximately **20.46%** of total revenue.

The **North region** generated the lowest revenue at approximately **₹49.03 million**, so there is an opportunity to investigate why its contribution is lower.

6. Product-category performance

Fashion is the strongest product category, generating **₹45,007,234**, approximately **17.90%** of total revenue.

Beauty generated the lowest category revenue at approximately **₹39.89 million**.

7. Processing efficiency

Average processing time is **36.55 hours**. This KPI should be monitored because reducing processing time can potentially improve delivery performance.

8. Customer satisfaction

The average customer rating is approximately **2.99 out of 5**, indicating that there is considerable room for improving customer experience.

2. KPI-Based Recommendations

Recommendation 1 — Improve On-Time Delivery

KPI: On-Time Delivery Rate = **74.72%**

The business should investigate the causes of the **17.16% delayed orders** and identify whether delays are concentrated around particular couriers, warehouses, regions, or priorities.

Action: Compare delivery performance by Courier, Warehouse, Region, and Priority.

Recommendation 2 — Reduce Cancellations

KPI: Cancellation Rate = **8.12%**

The 812 cancelled orders represent a meaningful loss of potential sales and customer activity.

Action: Analyze `Cancellation_Reason` to identify the major causes and address the most frequent ones.

Recommendation 3 — Optimize Logistics Cost

KPI: Total Shipping Cost = **₹2,758,648**

The average shipping cost is **₹275.86 per order**.

Action: Compare shipping costs across courier partners, regions, and priority levels to identify opportunities for cost reduction.

Recommendation 4 — Improve Processing Efficiency

KPI: Average Processing Time = **36.55 hours**

Action: Analyze processing time by warehouse, priority, and product category. High-processing areas should be investigated for workflow or inventory issues.

Recommendation 5 — Focus on High-Performing Categories

KPI: Revenue by Product Category

Fashion generated the highest revenue at **₹45.01 million**.

Action: Maintain sufficient inventory and marketing support for high-performing categories while investigating why lower-performing categories such as Beauty generate less revenue.

Recommendation 6 — Improve Customer Experience

KPI: Average Customer Rating = **2.99/5**

Action: Investigate customer feedback and connect ratings with delivery status, courier, product category, and processing time to identify the main causes of dissatisfaction.

3. Dataset Limitations

You can put this directly into your **README.md** or **project report**:

Dataset Limitations

1. The dataset contains **10,000 orders**, so the findings represent this dataset and may not represent the entire business population.
2. The data is historical and therefore may not accurately predict future customer behavior or operational performance.
3. External factors such as holidays, weather conditions, traffic, supply-chain disruptions, and market conditions are not included.
4. Customer demographic information is not available, limiting deeper customer segmentation.

5. Competitor pricing and competitor delivery performance are not available.
6. Dataset 2 contains missing values in some delivery-related fields, which need to be interpreted according to order status rather than automatically treated as errors.
7. The average customer rating of **2.99/5** provides a useful indicator, but additional customer feedback would be needed to understand the reasons behind the ratings.

4. Cleaning Log — Both Datasets

This is particularly important because your two datasets have different data-quality characteristics.

Cleaning Check	Dataset 1	Dataset2
Rows before cleaning	10,000	10,000
Columns	21	21
Duplicate rows	0	0
Missing cells	0	11,624
Rows after duplicate removal	10,000	10,000
Final rows	10,000	10,000

Important observation

Dataset 2 has missing values in:

Column	Missing Values
Actual_Delivery_Date	812
Delivery_Time_Days	812
Delay_Days	812
Cancellation_Reason	9,188
Total	11,624

Cleaning Actions Taken

You can document your cleaning process like this:

Cleaning Log

1. Loaded both datasets.
2. Verified row and column counts.
3. Checked for duplicate records.
4. Checked missing values.
5. Verified date columns.
6. Verified numeric columns such as Order_Value, Shipping_Cost and Processing_Time_Hours.
7. Checked Delivery_Status values.
8. Reviewed missing delivery fields against Cancellation status.
9. Confirmed that cancelled orders account for the missing delivery-related fields in Dataset2.
10. Retained all 10,000 records because there were no duplicate rows and the missing values were explainable by business context.

Validate the Business Logic of Missing Values

Missing Delivery Information for Cancelled Orders

Cancelled orders may not have delivery information because the order was never delivered. Therefore, missing values in Actual_Delivery_Date, Delivery_Time_Days, and Delay Days are expected for cancelled orders.

Cancellation Reason for Non-Cancelled Orders

For orders where Cancellation is "No", a missing Cancellation_Reason is expected because the order was not cancelled. This check confirms that there are no unexpected cancellation reasons recorded for non-cancelled orders.